



NITI Aayog



SECTOR AT A GLANCE 03

Wind Energy in India

Energy in India

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WIND ENERGY IN INDIA

ENERGY IN INDIA

The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

India has made significant progress in enhancing its wind capacity. Wind energy contributed to 34% of the cumulative installed power generation capacity of Renewable Energy Sources (RES) in June 2023.¹ India ranks as the fourth-largest market for onshore wind installations worldwide, contributing 5% to the total global onshore wind installations.

India possesses significant potential for both onshore and offshore wind energy. According to the India Energy Security Scenarios (IESS) 2047, wind energy is predicted to contribute approximately 257720.8 GWh to India's total energy supply by 2032, an increase from 68617 GWh in 2022.² The projected installation of 19 GW of onshore wind energy in India from 2022 to 2026 will generate 1 million direct and indirect jobs.³

Given the opportunities and challenges for promoting the deployment of offshore and onshore wind farms and wind-solar hybrids, workforce development, and establishing India as a manufacturing hub for wind turbines, the central government has supported several policy initiatives. These initiatives focus on expanding wind energy in India and include the National Offshore Wind Energy Policy, 2015, the National Wind-Solar Hybrid Policy, 2019 and the Vayumitra programme.

Promoting domestic manufacturing of critical components, repowering of old wind farms, supporting workforce development, and enabling regulatory and technological innovations to enhance grid integration and offshore capacity will be critical to continue advancing India's wind ambitions. This note presents key trends related to different components of wind energy in India and actions for the way forward.

Figure 1: Key wind energy statistics

43.7 GW

Installed Wind Energy Capacity in 2023

10%

Wind energy's contribution to India's installed generation capacity in 2023

50,000

Jobs created in the wind energy sector

97.02 Mtoe

Wind energy's projected contribution to India's total energy supply by 2047

99.9 GW

Estimated Installed Wind Energy Capacity by 2030

1 million

Direct and indirect jobs to be created by the wind energy sector by 2026

Source: NITI Aayog, India Energy Security Scenarios (IESS) 2047, 2023; Global Wind Energy Council, Accelerating Onshore Wind Capacity Addition in India to Achieve the 2030 Target, 2022; Ministry of New and Renewable Energy, Installed Wind Energy Capacity, 2023; Skills Council for Green Jobs, CEEW, NRDC, India's Expanding Clean Energy Workforce, 2023

¹ Central Electricity Authority, Executive Summary on Power Sector, June 2023; Note: RES include Small Hydro Project, Biomass Power, Urban and Industrial Waste Power, Solar and Wind Energy

² NITI Aayog, India Energy Security Scenarios (IESS) 2047, 2023

³ Global Wind Energy Council, Accelerating Onshore Wind Capacity Addition in India to Achieve the 2030 Target, 2022

Key trends

This section dives deeper into India's wind energy landscape, outlining statistics and trends on:

1. Resource planning and data management
2. Manufacturing, assembly, and standards
3. Generation, integration, and storage
4. Wind power utilisation
5. Workforce development
6. Waste management
7. Research and development
8. International collaboration
9. Monitoring and evaluation

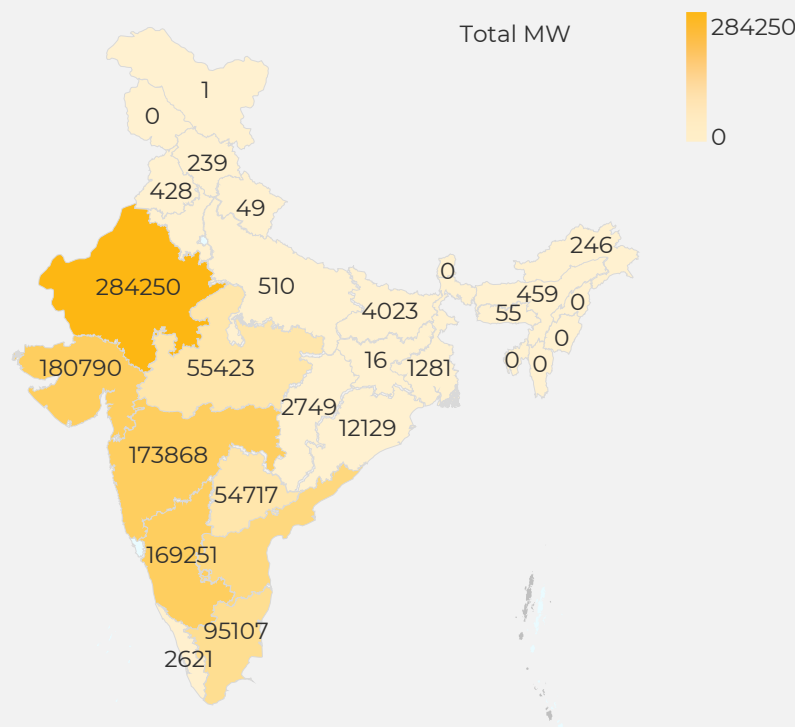
Resource planning and data management

India's wind energy planning and forecasting tools are robust, providing critical wind energy maps and data at different hub heights.

India's wind energy potential is more than 300 GW at 100m hub height⁴:

As of May 2022, only approximately 20% of the commercially utilisable potential had been leveraged.⁵ India possesses a technical resource of about 70 GW in offshore wind energy, primarily situated off the coasts of Gujarat and Tamil Nadu.⁶

Figure 2: Wind energy potential by state and UT at 150 metres above ground level (2023)



Source: National Institute of Wind Energy, Indian Wind Potential Atlas at 150m agl, 2023

⁴ National Institute of Wind Energy, Wind Power Potential at 100m agl, Accessed June 2023

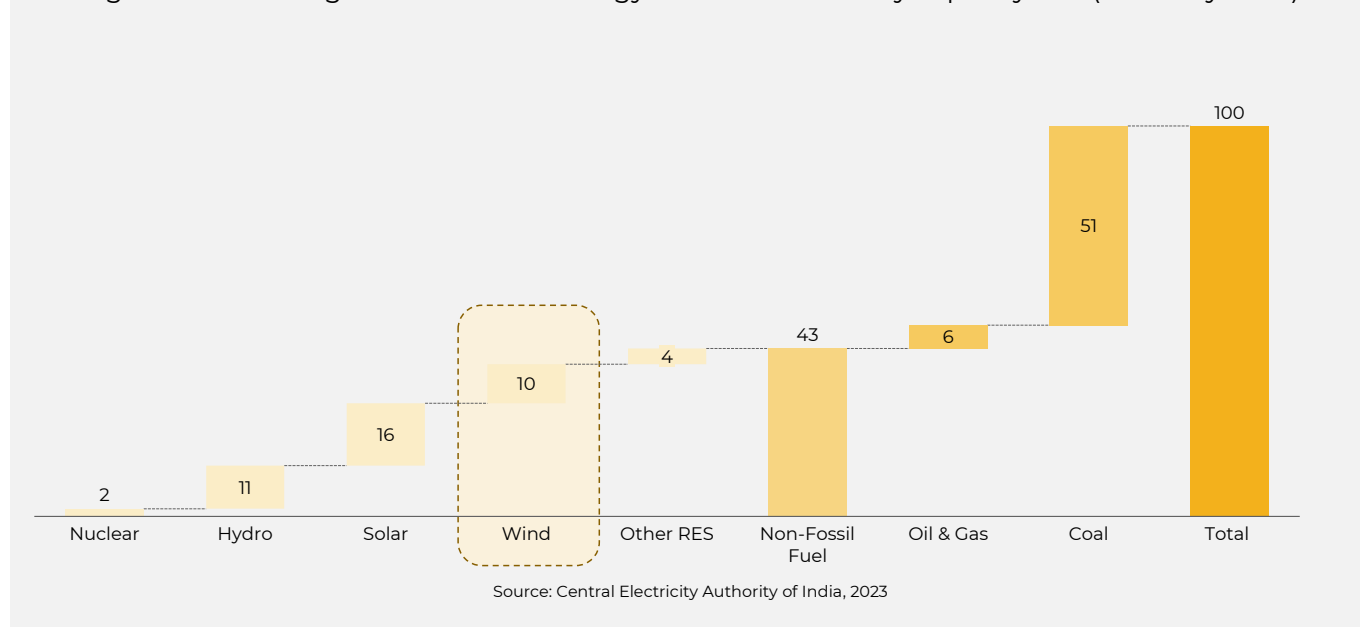
⁵ PRS Legislative Research, Standing Committee Report Summary - Evaluation of Wind Energy in India, 2022

⁶ Ministry of New and Renewable Energy, Offshore Wind, Accessed August 2023

The installed capacity of wind energy has increased significantly in the last decade:

Between March 2014 and June 2023, the installed capacity of wind power has experienced about 90% growth, while solar power has seen a much more significant increase of about 2000%.⁷ As of May 2023, wind power is a significant component of India's renewable energy mix, accounting for about 10% (equivalent to about 42 GW) of the total installed power generation capacity in India.⁸

Figure 3: Percentage share of wind energy in India's electricity capacity mix (February 2023)



Wind energy is expected to comprise 17% of India's energy mix by 2030:

As per the India Energy Security Scenarios (IESS) 2047, wind energy is projected to contribute approximately 22.16 (1.5%) Mtoe of the total energy supply in India by 2032, up from 5.9 Mtoe (0.6%) in 2022.⁹

Manufacturing, assembly, and standards

Manufacturing and assembly for wind energy in India includes producing turbine components like blades and towers and integrating these parts into functional wind turbines. It is essential to meet established quality and safety standards for efficient power generation.

Ministry of New and Renewable Energy (MNRE) is the governing body that sets standards for the wind sector in India:

The National Institute of Wind Energy (NIWE) collaborates with MNRE and sets guidelines and standards on topics such as micro siting, wind measurement, technical requirements for small and large wind turbine manufacturers, power quality in grid-connected wind energy, offshore wind assessments and environmental impact assessments, among others.¹⁰

India is the second-largest hub in the Asia-Pacific region for wind turbine manufacturing, contributing to 7% of the global wind turbine manufacturing capacity as of 2023¹¹:

India primarily leads in the assembly of wind turbines and the production of essential components such as gearboxes, blades, and generators in the APAC region.¹² The country's current annual capacity for

⁷ PRS Legislative Research, Standing Committee Report Summary - Evaluation of Wind Energy in India, 2022

⁸ Ministry of Power, Power Sector at a Glance, 2023

⁹ NITI Aayog, India Energy Security Scenarios (IESS) 2047, 2023

¹⁰ National Institute of Wind Energy, Documents, Accessed June 2023

¹¹ Global Wind Energy Council, Global Wind Report, 2023

¹² Global Wind Energy Council, Global Wind Report, 2023

manufacturing domestic wind turbines is approximately 15,000 MW.¹³ Notably, the wind sector has achieved significant indigenisation by increasing domestic manufacturing of components which were earlier being imported, with around 70-80% reliance on domestic manufacturing.¹⁴ Two Indian origin wind turbine manufacturers, namely Suzlon Energy Ltd. and Inox Wind make up a significant share of India's wind turbine supplier market (30%), with a 19% and 11% share respectively.¹⁵ However, some critical components such as balsa wood and pultruded carbon fiber are still being imported.

Mandates for using domestically produced wind components have helped to grow India's domestic manufacturing base:

Only equipment produced by manufacturers on the Revised List of Models and Manufacturers list can be utilised for wind energy projects, which has helped boost domestic manufacturing of wind components.¹⁶ Additionally, the government has made it mandatory for the assembly and manufacturing facility of the hub and nacelle, one of the four critical components of wind turbines along with blades and towers, to be located in India.¹⁷

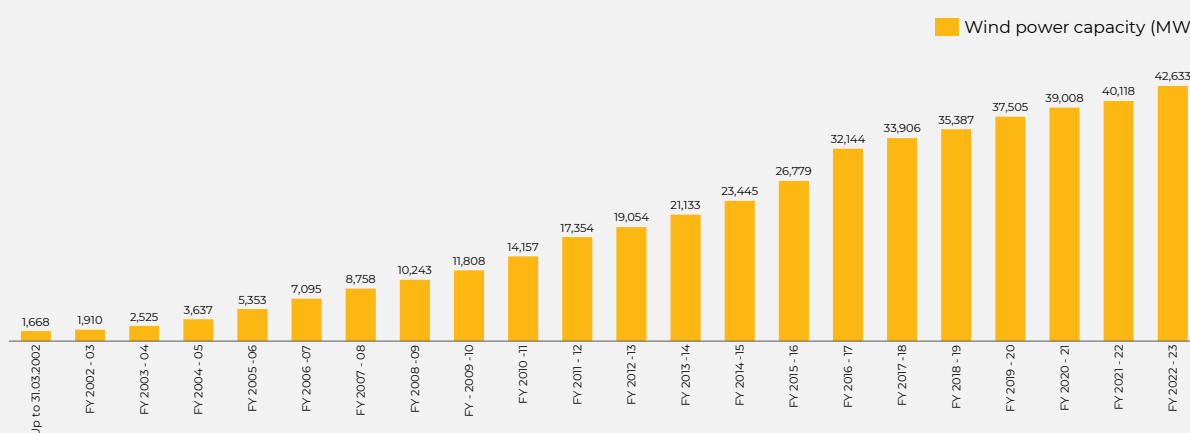
Generation, integration, and storage

Generation of wind energy involves harnessing wind power through turbines, integrating this electricity into the grid for nationwide distribution, and exploring storage solutions like batteries to manage intermittent power supply efficiently.

Growth rate for wind energy has declined in the past five years:

In the past five years, the rate of wind energy installations has decreased, with an average growth rate of about 5% between 2019-2023, compared to around 12% between 2012-2016.¹⁸ A slowdown was observed during 2017-2019, which can be attributed to multiple changes in wind policies, such as the introduction of reverse bidding.¹⁹ The slowdown in 2020-2021 can be attributed to the COVID-19 pandemic's first and second waves and disruptions in the wind supply chain during the monsoon season.²⁰

Figure 4: Cumulative growth of wind power capacity in India (MW)



Source: National Institute of Wind Energy, Indian Wind Potential Atlas at 150m agl, 2023

¹³ Ministry of New and Renewable Energy, Wind Overview, Accessed June 2023

¹⁴ Ministry of New and Renewable Energy, Wind Energy – Manufacturing, Accessed June 2023

¹⁵ Bloomberg NEF, India's Top Wind Suppliers in 2019, 2020

¹⁶ Ministry of New and Renewable Energy, Government incentivizes local development and manufacturing of renewable energy technologies, 2022

¹⁷ Ministry of New and Renewable Energy, Wind Energy – Manufacturing, Accessed June 2023

¹⁸ Central Electricity Authority, All India Electricity Statistics, 2023; Ministry of Power, Power Sector at a Glance, 2023

¹⁹ Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

²⁰ Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

Policy efforts are focusing increasingly on wind-solar hybrid projects:

The objective of the National Wind-Solar Hybrid Policy, 2018, is to encourage the utilisation of large-scale grid-connected wind-solar PV hybrid systems to optimise the use of wind and solar energy resources, transmission infrastructure, and land. The policy aims to maximise the effectiveness and efficiency of harnessing both wind and solar energy in a coordinated manner.²¹

MNRE has established several renewable energy policy incentives that benefit the wind energy sector:

In November 2021, an Inter-State Transmission System (ISTS) waiver extension was announced, which waived off ISTS charges for renewable energy (RE) projects commissioned before June 2025 for 25 years, which helps reduce costs and incentivises the development of wind power projects. A May 2022 policy announcement allowed commercial and industrial (C&I) consumers and distribution companies to trade excess Renewable Energy Certificates (RECs), allowing for greater flexibility in utilising renewable energy resources, including wind. In June 2022, the Green Open Access Rules were announced, making the approval process for RE projects, including wind, easier while providing additional support and benefits projects.²²

Wind power utilisation

Demand for wind power is increasing across India, largely being driven by central mandates for renewable energy.

The Renewable Purchase Obligation (RPO) is a significant factor driving the demand for wind installation:

Central auction procurement is primarily driven by demand from 14 states without abundant wind resources and eight states with ample wind resources.²³ According to India's Wind Energy Market Outlook 2022-2026 released by GWEC, the state distribution companies (DISCOMs) in four windy states, namely Karnataka, Andhra Pradesh, Gujarat, and Maharashtra, are expected to conduct their own auctions to procure power, reflecting the unmet demand in those regions.²⁴

Waivers have helped increase the demand for wind projects in India:

Gujarat, Rajasthan, and Andhra Pradesh have introduced hybrid policies that include exemptions from Open Access (OA) charges for wind-solar projects. Furthermore, the central government's announcement of Inter-State Transmission System (ISTS) waivers until 2025 is already increasing the adoption of hybrid installations by the Commercial and Industrial (C&I) segment, as the Tranche IV Hybrid auctions were oversubscribed by around five times the awarded capacity in 2021.²⁵

Workforce development

The wind sector in India has created about 50,000 employment opportunities.²⁶ Workforce development is crucial to expand wind energy by fostering skilled labour capable of operating, maintaining, and innovating wind technology.

Wind sector to create 1 million jobs in the next three decades:

Over 25 years, the projected installation of 19.4 GW of onshore wind energy in India from 2022 to 2026 is anticipated to generate 1 million direct and indirect jobs. These jobs will span various sectors, including

²¹ Ministry of New & Renewable Energy, National Wind-Solar Hybrid Policy, 2018

²² Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

²³ National Institute of Wind Energy, India's Wind Potential Atlas at 120m agl, 2019; Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

²⁴ Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

²⁵ Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

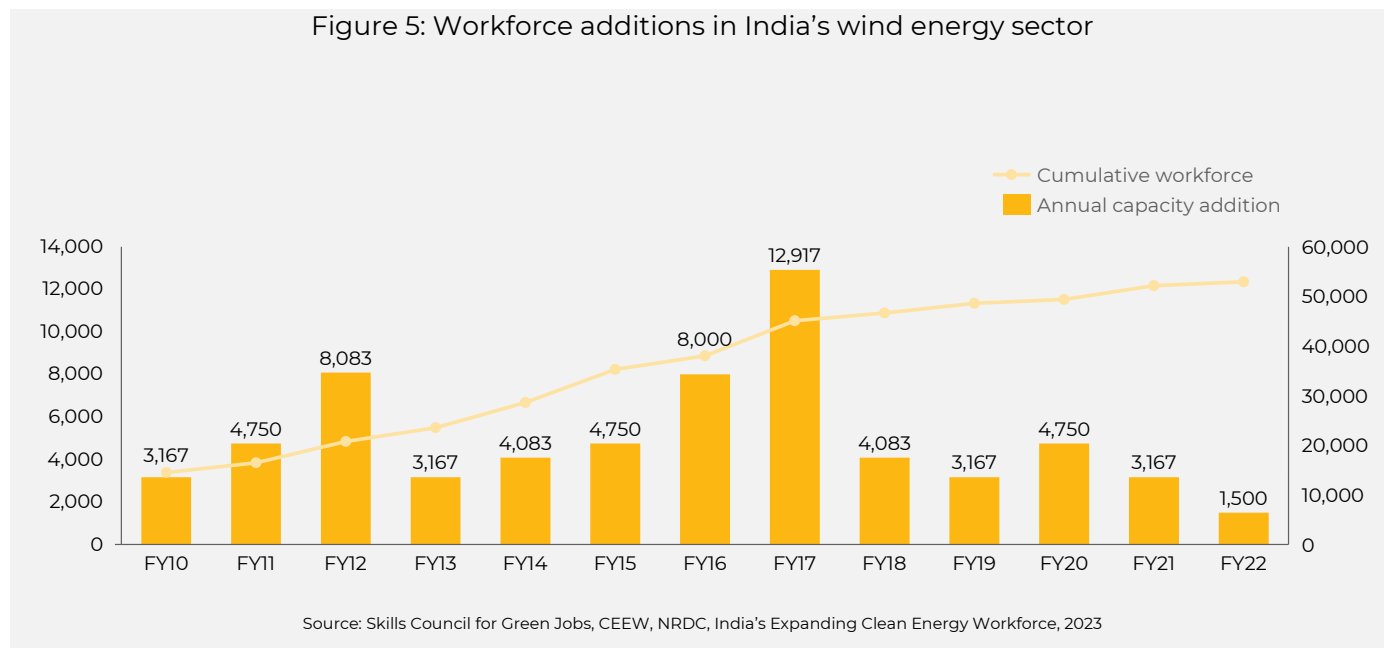
²⁶ Skills Council for Green Jobs, CEEW, NRDC, India's Expanding Clean Energy Workforce, 2023

manufacturing, project development, construction, operations, and maintenance (O&M), and installation.²⁷

Recent reforms are expected to support job creation:

The recent changes in the auction process for wind projects have eliminated the reverse bidding mechanism, which was found to be unsustainable for developers. Moreover, the government has established a target of inviting bids for 8 GW of wind energy projects annually until 2030. These measures are anticipated to rejuvenate capacity deployment in the sector and create new job opportunities.²⁸

Figure 5: Workforce additions in India's wind energy sector



Waste management

Effective waste management in wind energy includes handling decommissioned turbine components responsibly and ensuring sustainable disposal methods.

Wind energy is estimated to generate 1.1 million tonnes of waste by 2050:

Typically, blades and nacelles are decommissioned after 20-25 years. These decommissioned components are made of inert reinforced polymers that provide structural stability and strength. However, because they are composed of thermoset polymers, they cannot readily degrade in landfills and thus require specific waste treatment. Similarly, while metals from wind turbines can be recycled easily, other materials like paints, adhesives, and composite materials pose challenges for recycling. Wind energy waste volume is expected to peak in 2037, primarily due to the installation of 5502 MW of power.²⁹

Guidelines for recycling some wind turbine components exist:

Most components of wind turbines are composed of recyclable metals, and for the Fiber Reinforced Plastics (FRP) utilized in blades, the Central Pollution Control Board (CPCB) released 'Guidelines for Disposal of Thermoset Plastic Waste including Sheet moulding compound (SMC) / Fiber Reinforced Plastic (FRP)' in 2016.³⁰

²⁷ Global Wind Energy Council, Accelerating Onshore Wind Capacity Addition in India to Achieve the 2030 Target, 2022

²⁸ Council on Energy, Environment and Water (CEEW), Report on India's Expanding Clean Energy Workforce, 2023

²⁹ Housing and Urban Development Corporation Ltd (HUDCO), Wind energy in India - Waste generation and end-of-life management for sustainability, 2021

³⁰ MNRE, Wind Power Potential of Windy States and Policy for Repowering Wind Power Projects, 2023

Research & Development

Several ongoing R&D efforts seek to support resource assessment and technology deployment for wind energy.

India's R&D efforts on wind energy focus on developing next-generation wind technologies and lowering the cost of wind energy projects:

The NIWE finds that India's R&D efforts in the wind sector may span efforts to address the key drivers of LCOE and overcome non-technical market barriers that hinder deployment. These include research efforts to characterise better and assess wind resources, develop new technologies, improve grid integration, and address social and environmental issues.³¹

Between 2012-16, R&D expenditure on wind energy was 3% of the total expenditure on R&D for renewable energy:

In 2015-16, India spent INR 3 crores on wind-hybrid energy R&D, which constitutes only 8% of the expenditure on solar energy R&D (INR 33 crores), and 3% of the total expenditure on renewable energy R&D that year.³²

International collaboration

India is a part of key renewable energy alliances and has entered multiple bilateral agreements to enhance domestic wind energy capabilities.

India's partnerships with international institutions and other countries include wind energy as a focus area:

Bilateral partnerships with United Kingdom focus on innovation and research and development, market and supply chain growth, testing and demonstration of new technologies, skills development programs, blade test facilities, and turbine and grid simulations, via programs like ASPIRE by FCDO, MoP, and MNRE and the Offshore Renewable Energy Catapult and NIWE collaborations.³³ India and Denmark launched a joint initiative to set up a knowledge hub called the Centre of Excellence for Offshore Wind and Renewable Energy (CoE).³⁴ The platform aims to bring together public civil society and public authorities and facilitate the implementation of the Indian offshore wind strategy.

India has entered multiple cooperation agreements to improve its wind energy capabilities:

The First Offshore Wind Project of India (FOWPI) 2016-19, led by COWI and supported by the EU, assisted in capacity-building of Indian stakeholders for harnessing offshore wind energy.³⁵ Additionally, the Department for Business and Trade India, a UK Government department, signed a strategic partnership with GWEC India in 2023 to support India's offshore wind goals through targeted interventions and knowledge exchange.³⁶ Furthermore, India signed a strategic sector cooperation agreement with the Ministry of Energy, Utilities and Climate of the Kingdom of Denmark in 2022 aimed at technical capacity building for managing offshore wind projects in India.³⁷

Monitoring and evaluation

³¹ National Institute of Wind Energy, White Paper On Wind Turbine Technology R&D For India 2020-2030, 2021

³² Ministry of New and Renewable Energy, Funds for R&D of New and Renewable Energy Technology, 2016

³³ Global Wind Energy Council, Recommendations for Offshore Wind Development in India, 2023

³⁴ Ministry of New and Renewable Energy, The Centre of Excellence for Offshore Wind and Renewable Energy, Accessed August 2023

³⁵ Global Wind Energy Council, Recommendations for Offshore Wind Development in India, 2023

³⁶ Global Wind Energy Council, Recommendations for Offshore Wind Development in India, 2023

³⁷ Ministry of New and Renewable Energy, Ministry of New and Renewable Energy signs a strategic sector cooperation agreement and letter of intent with Denmark to explore the sources of wind energy, 2022

The Ministry of New and Renewable Energy is the principal monitoring and evaluation body for wind energy projects in India, with support from the National Institute for Wind Energy (NIWE). In addition to NIWE, various specialised committees are tasked with overseeing offshore energy initiatives.

The National Offshore Wind Policy outlines the role of NIWE and the Offshore Wind Energy Steering Committee (OWESC) in monitoring progress:

The Offshore Wind Energy Steering Committee (OWESC) oversees the progress of offshore wind energy development in the country and is also tasked with providing policy guidance and ensuring the successful implementation of specific activities related to offshore wind energy. NIWE monitors developing, allocating, and constructing new offshore wind energy blocks. Coastal states' designated agencies also keep track of the development along their respective coastlines to plan for logistics and grid infrastructure.³⁸

³⁸ Ministry of New and Renewable Energy, Notification on National Offshore Wind Energy Policy, 2015

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by increasing the adoption of wind energy. These revolve primarily around the potential for decarbonisation, offshore wind energy, wind energy export and solar-wind hybrids to unlock other renewable sources of energy like Green Hydrogen.



Accelerate India's decarbonisation and RE transition:

India possesses a substantial onshore wind potential of 302 GW at a height of 100m, 695 GW at a height of 120m and 1164 GW at 150m.³⁹ Considering its cost-competitiveness and ability to complement other energy sources, such as solar, to provide round-the-clock solutions, wind energy will play a crucial role in India's ongoing transition in the power sector.⁴⁰



Harness offshore wind energy to increase wind's RE contribution:

Offshore wind energy achieves a greater capacity utilisation factor compared to onshore projects, and as the installed capacity rises, its cost diminishes. Regarding offshore wind, India has the potential of about 70 GW primarily located off the coasts of Gujarat and Tamil Nadu.⁴¹



Repower old wind farms for efficient capacity utilisation:

Most wind farms in India are located at the windiest sites but are old and working at less than their optimal capacity and potential.⁴² Power generation capacity and efficiency can be improved by substituting outdated components with newer, more powerful turbines that offer increased durability against environmental factors and upgraded materials.⁴³



Position India as a global wind component export hub:

India has already established itself as the world's second-largest hub for wind supply chain manufacturing, and further expansion in this sector will lead to additional job opportunities, bolster GDP growth, and foster technical advancements. This aligns with the government's "Make in India" initiative, promoting domestic manufacturing capabilities and reinforcing the country's position as a key player in the global wind energy market.⁴⁴



Skilling programs to enable workforce transition to green jobs:

The projected installation of 19 GW of onshore wind energy in India from 2022 to 2026 can generate 1 million direct and indirect jobs.⁴⁵ These jobs will span various sectors, including manufacturing, project development, construction, operations, and maintenance (O&M), and installation.⁴⁶ Thus, building the capacity of vocational and higher education institutions in wind energy-related curricular structures can aid the workforce transition.



Enable green hydrogen:

39 National Institute of Wind Energy, Wind Power Potential at 100m agl, Accessed July 2023; National Institute of Wind Energy, India's Wind Potential Atlas at 120m agl, 2019; National Institute of Wind Energy, India's Wind Potential Atlas at 150m agl, 2023

40 Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

41 Ministry of New and Renewable Energy, Offshore Wind, Accessed August 2023

42 Council on Energy, Environment and Water (CEEW), Wind Repowering, 2022

43 Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

44 Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

45 Global Wind Energy Council, Accelerating Onshore Wind Capacity Addition in India to Achieve the 2030 Target, 2022

46 Global Wind Energy Council, Accelerating Onshore Wind Capacity Addition in India to Achieve the 2030 Target, 2022

Solar-wind hybrids are critical to enable other RE sectors, primarily Green Hydrogen (GH2).⁴⁷ As complementary sources of energy, solar and wind can power GH2 electrolyzers through the day and night, respectively, making them the best suited to achieve GH2 production in India at scale.

Challenges

This section highlights some of the major challenges and constraints facing wind energy in India. Efforts to resolve such challenges will ensure availability of wind data, reduces cost of manufacturing turbines, workforce development and adequate repowering and recycling of wind turbines.



Limited assessment of offshore and urban wind data:

Limited mapping of exclusion zones and areas reserved for different ocean activities prevents developers from obtaining timely consent for offshore wind projects.⁴⁸ Further, currently, there is limited research on lower-level wind resources (10- to 20-meter height range) in urban areas. This research is vital to accurately assess wind potential and guide the long-term strategy for the development of indigenous wind turbines suitable for urban areas.⁴⁹ The MNRE has released three models to ramp up assessments of offshore wind energy in 2022.⁵⁰ Under these models, studies/ surveys will be conducted by prospective developers, states, NIWE or other government entities.



High-cost pressures on domestic equipment manufacturers:

Foreign and domestic original equipment manufacturers (OEMs) of wind turbines are adversely affected by growing costs of raw materials and critical wind turbine components like steel and nickel.⁵¹ The cost per megawatt for offshore wind turbines is two to three times higher than onshore wind turbines, significantly reducing their economic viability.⁵² A recent Goods and Services Tax (GST) surge has further increased turbine costs.⁵³ Further, though the cost of battery packs has been gradually declining at a CAGR of approximately 8% since 2018, the current high costs in 2022 present a hindrance to the expansion of wind-solar-hybrid (WSH) systems as a viable and sustainable solution for round-the-clock power supply.⁵⁴ A related challenge involves the need for greater local manufacturing of dual-function grid inverters to help reduce system costs (a shared roadblock to greater adoption of solar and wind energy solutions).⁵⁵



Limited policy support for repowering, land acquisition and recycling:

In India, most wind energy projects established before 2000 were strategically located in highly favourable wind locations to maximise output. However, these projects operate at a capacity below 500 kW and their maximum achievable plant load factors (PLF) remain below 20%, indicating suboptimal utilisation of their full potential.⁵⁶ The Draft National Repowering Policy for Wind Power Projects, 2022 has been released, however it is currently in its stakeholder consultation phase.⁵⁷ Further, land acquisition and title-clearing remain persistent obstacles for most onshore wind installations. Currently, the process of allocating land and obtaining permits for land use conversion takes approximately 6-9 months and sometimes up to 18-24 months.⁵⁸ Finally, while the recycling and end-of-life planning of wind

47 CEEW, A Green Hydrogen Economy for India: Policy and Technology Imperatives to Lower Production Cost, 2020

48 Global Wind Energy Council, Recommendations for Offshore Wind Development in India, 2021

49 NITI Aayog, Report on India's Renewable Electricity Roadmap 2030, 2015

50 Ministry of New and Renewable Energy, Offshore Wind, Accessed Augusts 2023

51 Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

52 Lok Sabha Standing Committee on Energy, Action Plan for Achievement Of 175 Gigawatt (Gw) Renewable Energy Target, 2021

53 Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

54 Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

55 TERI, Reaching the sun with rooftop solar, 2014

56 Council on Energy, Environment and Water (CEEW), Wind Repowering, 2022

57 MNRE, Draft National Repowering Policy for Wind Power Projects, 2022

58 Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

turbines in India has been highlighted as an R&D priority and certain guidelines for recycling exist, no official policy mandates recycling in the wind sector.⁵⁹



Inadequate workforce development:

The deployment of 1 GW of wind capacity resulted in the employment of 556 workers in 2022, which marks a 28% decline compared to the previous fiscal year. These workers were primarily involved in operation and maintenance activities, with no new job requirements arising for other project deployment phases.⁶⁰ Although the Skill Council for Green Jobs offers training programs, there are only six qualifications available for operations and maintenance, and no qualifications are provided for wind energy manufacturing.⁶¹ Under the Vayumitra training programme, only 1000 people have been trained so far.



Limited number of states that have align their trajectories with the updated Renewable Purchase Obligations (RPO):

Among the 17 states that procure power through central auctions, Odisha, Haryana, Uttar Pradesh, Madhya Pradesh, and Rajasthan have established their Renewable Purchase Obligation (RPO) trajectories, including wind energy, until 2026.⁶² However, the remaining states' trajectories are not defined beyond 2022, posing a risk of failing to meet the renewable energy targets, including wind, set in India.⁶³



Issues in grid integration of offshore wind energy:

India's current grid planning does not adequately account for the integration of offshore wind energy into the grid, causing delays in reinforcing the onshore grid.⁶⁴ Additionally, acquiring the required permissions and pathways for onshore transmission infrastructure faces hurdles, obstructing the progress of onshore wind energy projects.⁶⁵ The current grid codes also do not specifically address the unique characteristics of offshore wind and its transmission, such as the natural patterns in wind resources across geographies.⁶⁶ Relatedly, efforts such as those outlined by the CEA's cyber security guidelines remain necessary to ensure cyber resilience given the increasing use of connectivity technologies in power systems.⁶⁷

59 National Institute of Wind Energy, White Paper On Wind Turbine Technology R&D For India 2020-2030, 2021

60 Council on Energy, Environment and Water (CEEW), Report on India's Expanding Clean Energy Workforce, 2023

61 Skills Council for Green Jobs, Gearing Up the Indian Workforce for a Green Economy, 2023

62 Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

63 Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

64 FOWIND Consortium, Grid Integration Study for Offshore Wind Farm Development in Gujarat and Tamil Nadu, 2017

65 FOWIND Consortium, Grid Integration Study for Offshore Wind Farm Development in Gujarat and Tamil Nadu, 2017

66 FOWIND Consortium, Grid Integration Study for Offshore Wind Farm Development in Gujarat and Tamil Nadu, 2017

67 CEA, Cyber Security in Power Sector Guidelines, 2021

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee wind energy.



NITI Aayog

The NITI Aayog serves as the apex public policy think tank of the Government of India. Within the energy vertical, it deals with five key sectors: power, coal, petroleum and natural gas, new and renewable energy, and atomic power. Conducts policy analysis of the wind energy sector; and undertakes steps to improve energy planning, and data management in India



Ministry of Power

Administers, formulates and implements power sector-related policies and provides R&D and technical assistance relating to grid integration, transmission system networks and distribution systems in the States/UTs



Ministry of New and Renewable Energy (MNRE)

Develops renewable and clean energy technologies, systems, products, and services in line with global standards and deploys these for energy applications for sectors like power, transport and support country's objective of achieving energy security



Bureau of Energy Efficiency

Assists in developing policies and strategies to promote energy efficiency and conservation and utilises existing resources and infrastructure in performing functions under the Energy Conservation Act



Central Electricity Authority

Acts as a technical support base for power sector stakeholders and supports policy formulation, establishing technical standards, project monitoring and human resource skilling



Central Electricity Regulatory Commission

Improves regional transmission system operations, formulates efficient tariff mechanisms, promotes power market development, and facilitates technological advancements alongside environmental and safety compliance



National Institute of Wind Energy

Engages in research and development related to wind energy, conduct wind resource assessment (both onshore and offshore), establishes standards, and carries out testing, certification, and skill development initiatives for the wind energy sector

Major policies and programs

This section highlights existing policies related to wind energy and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



National Offshore Wind Energy Policy, 2015⁶⁸

The Policy aims to establish offshore wind power projects through:

- Promoting the deployment of offshore wind farms in the Exclusive Economic Zones (EEZ) of the country
- Encouraging spatial planning and management of maritime renewable energy resources in the exclusive economic zone of the country through suitable incentives



National Wind Solar Hybrid Policy, 2018⁶⁹

The Policy aims to achieve hybridisation of existing wind and solar projects through:

- Providing a framework for the promotion of large grid-connected wind-solar PV hybrid systems for efficient utilisation of transmission infrastructure and land

Encouraging new technologies, and methods involving the combined operation of wind and solar PV plants



CEA (Cyber Security in Power Sector) Guidelines, 2021⁷⁰

Guidelines released by the Central Electricity Authority to combat the cybersecurity threats facing power supply systems and grid operations in India.



Green Energy Open Access Rules, 2022⁷¹

The rules aim to promote the generation, purchase, and consumption of green energy through an open access registry. The registry serves as a transparent platform overseeing and executing long, medium, and short-term open access transactions within both inter-state and intra-state transmission systems.



Green Energy Corridor Scheme⁷²

The Green Energy Corridor scheme aims to facilitate the evacuation and integration of renewable energy. It involves the developing transmission systems and control infrastructure to increase India's overall RE capacity by harnessing the potential of RE rich states like Andhra Pradesh, Gujarat, Himachal Pradesh, Jammu and Kashmir, Karnataka, Maharashtra, Rajasthan, Madhya Pradesh, and Tamil Nadu.

⁶⁸ Ministry of New and Renewable Energy, Wind Policies, 2015

⁶⁹ Ministry of New and Renewable Energy, National Wind-Solar Hybrid Policy, 2018

⁷⁰ Central Electricity Authority, CEA (Cyber Security in Power Sector) Guidelines, 2021

⁷¹ GRID India – National Load Dispatch Centre, Green Energy Open Access Registry, Accessed December 2023

⁷² Ministry of Power, Transmission Works under Green Energy Corridors-I, Accessed December 2023



Central Electricity Regulatory Commission (Indian Electricity Grid Code) Regulations, 2010

The guidelines mandate the “must-run” status of all wind energy plants.



Vayumitra Skill Development Programme⁷³

Run by the National Institute of Wind Energy (NIWE), the programme aims to develop a proficient workforce tailored for India's wind energy sector. It conducts trainings in nine windy states for positions such as O&M Electrical & Instrumentation Technician, O&M Mechanical Technician, and site surveyor positions. The programme is designed in accordance with the National Skill Development Corporation (NSDC) of the Ministry of Skill Development and Entrepreneurship, Government of India.



Additional policies and provisions to promote wind energy in India

- Reverse auctions for wind projects have been eliminated, replaced by a traditional bid system comprising two parts: technical and financial evaluations.
- SECI (Solar Energy Corporation of India) will solicit bids for up to 4GW capacity to establish offshore wind farms along the coasts of Tamil Nadu and Gujarat, supporting offshore wind initiatives⁷⁴
- The updated Renewable Purchase Obligation (RPO) compliance facilitates integrating wind power up to 6.94% of the electricity acquired by DISCOMs/states until 2029-30⁷⁵
- The draft National Repowering Policy for wind power projects has been released to optimise the utilisation of wind energy resources by maximizing energy yield (kWh) per square kilometre in wind project areas⁷⁶
- Certain components essential for manufacturing wind electric generators are granted exemption from concessional custom duties⁷⁷
- Wind projects commissioned on or before March 31, 2017, receive Generation Based Incentives (GBI) as a support measure⁷⁸
- The National Institute of Wind Energy in Chennai offers technical assistance such as wind resource assessment and identification of potential sites⁷⁹
- Foreign Direct Investment (FDI) up to 100% is allowed through the automatic route⁸⁰
- Inter State Transmission System (ISTS) charges have been waived for cross-state sales of solar and wind power for projects to be commissioned by June 30, 2025⁸¹
- Ultra Mega Renewable Energy Parks are being established to offer land and transmission infrastructure to renewable energy developers for large-scale project installations⁸²

⁷³ NIWE, Vayumitra Skill Development Programme, Accessed December 2023

⁷⁴ Ministry of Power, Offshore Wind Energy in India, 2022

⁷⁵ Ministry of Power, RPO Obligation Order, 2022

⁷⁶ Ministry of New and Renewable Energy, National Repowering Policy for Wind Power Projects – Draft, 2022

⁷⁷ Ministry of New and Renewable Energy, Steps by Government of India to promote renewable energy in the country, 2022

⁷⁸ Ministry of New and Renewable Energy, Steps by Government of India to promote renewable energy in the country, 2022

⁷⁹ Ministry of New and Renewable Energy, Steps by Government of India to promote renewable energy in the country, 2022

⁸⁰ Ministry of New and Renewable Energy, Steps by Government of India to promote renewable energy in the country, 2022

⁸¹ Ministry of New and Renewable Energy, Steps by Government of India to promote renewable energy in the country, 2022

⁸² Ministry of New and Renewable Energy, Steps by Government of India to promote renewable energy in the country, 2022

- New transmission lines are being constructed and sub-station capacity being expanded to accommodate the transmission of renewable energy⁸³
- A Project Development Cell is being established to attract and assist investments⁸⁴
- Standard Bidding Guidelines have been implemented for competitive tariff-based bidding in procuring power from grid connected wind projects⁸⁵
- Power dispatch has been mandated upon receipt of a Letter of Credit (LC) or advance payment by distribution licensees to ensure punctual payment to renewable energy generators⁸⁶

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83 Ministry of New and Renewable Energy, Steps by Government of India to promote renewable energy in the country, 2022

84 Ministry of New and Renewable Energy, Steps by Government of India to promote renewable energy in the country, 2022

85 Ministry of New and Renewable Energy, Steps by Government of India to promote renewable energy in the country, 2022

86 Ministry of New and Renewable Energy, Steps by Government of India to promote renewable energy in the country, 2022

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. These are based on reports and materials prepared by key government bodies and expert institutions like the Global Wind Energy Council, the Skills Council for Green Jobs, and CEEW.

 Boost domestic manufacturing	Provide state-level policy support for original equipment manufacturers⁸⁷ - Promote state-level measures such as production-linked incentives (PLI) that are linked to commissioned capacity - Provide concessions for components that have to be mandatorily imported	Enhance wind energy skilling courses⁸⁸ - Increase the availability of wind energy skilling courses in all states - Ensure accessibility to these courses in rural and coastal areas to prepare a specialised workforce for potential offshore wind energy projects - Offer courses across the wind energy value chain, including installation, maintenance, and operation
 Enhance technology deployment and upgrades	Speed up the repowering of old wind projects through state-level initiatives⁸⁹ - Proactively assess the annual volume of wind capacity nearing the end of its lifetime - Enable repowering via state-level measures that fast-track regulatory requirements - Design win-win financial models that provide incentives for all stakeholders involved in repowering to lower upfront costs - Deploy state-level policies mandating the safe disposal, and wherever relevant, the reuse of material and equipment from wind farms	Provide additional incentives for OEMs for technological upgrades⁹⁰ - Offer tax credits or financial incentives to OEMs that invest in research and development for technological upgrades in wind energy equipment - Establish grant programs that promote collaboration between OEMs and research institutions
	Expand pilot testing for offshore wind energy and micro wind farms	Enhance policy planning for grid integration of offshore wind energy⁹¹

⁸⁷ Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

⁸⁸ Skills Council for Green Jobs, Gearing Up the Indian Workforce for a Green Economy, 2023

⁸⁹ Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

⁹⁰ Global Wind Energy Council, India Wind Energy Market Outlook 2022-2026, 2022

⁹¹ CEEW, Assessing the Value of Offshore Wind for India's Power System in 2030, 2023; FOWIND Consortium, Grid Integration Study for Offshore Wind Farm Development in Gujarat and Tamil Nadu, 2017; NITI Aayog and IEA, Renewables Integration in India, 2021

Continue promoting innovation, R&D, and grid integration	• Broaden pilot testing initiatives to diverse coastal regions and potential offshore sites • Explore micro wind farms in inland areas to assess their efficacy and suitability	• Conduct financial assessments for establishing evacuation infrastructure for future offshore wind energy projects • Create regulatory frameworks outlining clear guidelines for the seamless integration of offshore wind energy into the national grid, including through efforts to create a new ancillary services market, upgrading state grid codes to enable system-friendly integration, and promoting a financial market that makes trading power sector products possible • Define technical standards and specifications for offshore connections, ensuring compatibility and efficient transmission of electricity from offshore wind farms to onshore grids.
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Recommended readings

Skills Council for Green Jobs. (2023). *Gearing Up the Indian Workforce for a Green Economy*.
<https://sscgj.in/wp-content/uploads/2023/05/Skills-Landscape-for-Green-Jobs-Report.pdf>

Global Wind Energy Council. (2022). *India Wind Energy Market Outlook 2022-2026*.
<https://gwec.net/wp-content/uploads/2022/08/India-Outlook-2026.pdf>

Ministry of New and Renewable Energy. (2021). *Action Plan for Achievement of 175 Gigawatt (GW) Renewable Energy Target*.
https://loksabhadocs.nic.in/lsscommittee/Energy/17_Energy_17.pdf

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